

Day 15 Student Handout

Geometry Summer Credit | Right-Triangle Trig Synthesis

Name:	Date:	Period:
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Learning Targets

I can choose an efficient right-triangle strategy. I can find triangle area using two sides and the included angle. I can explain and check a complete geometric model before the weekly assessment.

Lesson 15A: Choose the Strategy

Warm-Up

1.	Find the hypotenuse of a right triangle with legs 8 and 15.
2.	A 45-45-90 triangle has leg 6. Find the hypotenuse exactly.
3.	A right triangle has angle 37 degrees and adjacent side 12. Find the opposite side.

Strategy Map

Match the Information to the Tool

Two sides of a right triangle: Pythagorean theorem. A 45-45-90 or 30-60-90 triangle: special ratios. An acute angle plus a side, or two sides plus an unknown angle: sine, cosine, or tangent. Two sides and their included angle: $A = (1/2)ab \sin(C)$.

Problem	Work / notes
A right triangle has legs 9 and 40. Find the hypotenuse.	Best tool: Setup: Answer:
A 30-60-90 triangle has hypotenuse 18. Find both legs.	Best tool: Exact answers:
A right triangle has angle 52 degrees and hypotenuse 20. Find the adjacent leg.	Best tool and ratio: Equation: Answer:

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Lesson 15B | Trig Area and Weekly Review

Area Using Trigonometry

Two Sides and the Included Angle

If sides a and b form included angle C , then $A = (1/2)ab \sin(C)$. The sine factor gives the perpendicular-height portion of one side. The angle must be between the two given sides.

Problem	Work / notes
Two sides of a triangle are 10 cm and 14 cm with included angle 38 degrees. Find the area.	Formula: Substitute: Area with units:
A triangular garden has sides 18 m and 25 m with included angle 64 degrees. Find its area.	Formula: Substitute: Area with units:
Explain why $(1/2)ab \sin(C)$ becomes $(1/2)ab$ when $C = 90$ degrees.	$\sin(90) =$ Connection to base-height area:

Optional Enrichment - Not Assessed

Law of Sines and Law of Cosines

These tools solve non-right triangles and are enrichment only. Law of Sines: $a/\sin(A) = b/\sin(B) = c/\sin(C)$. Law of Cosines: $c^2 = a^2 + b^2 - 2ab \cos(C)$. They will not appear on today's weekly assessment.

Assessment Readiness

4.	Name the best first strategy for each: (a) two legs known, (b) a 30-60-90 triangle, (c) angle and hypotenuse known, (d) two sides and their included angle for area.
5.	Write one calculator or reasonableness check you will use today.